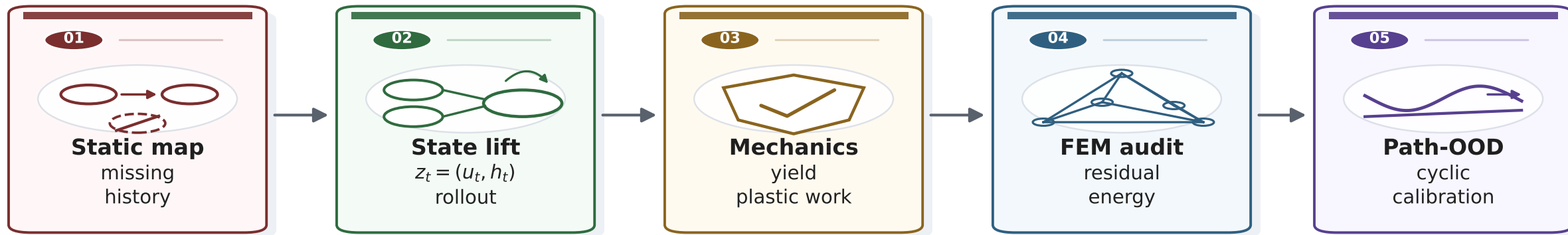


Theory chain

from static surrogate failure to stateful, mechanics-audited prediction



Evidence chain

field

history

residual

energy

yield/work

reversal

calibration